Course Code: BTCS 606-18	**Course Title:** Simulation and Modeling	**3L:0T:0P**	**3Credits**

Detailed Contents:

UNIT1: Introduction

Introduction to simulation and modeling, Application areas, System and system environment, Components of a system, Discrete and continuous systems, Basic model forms and its types, Discrete-event simulation, Steps in a simulation study, Simulation examples.**[4 hrs] (CO 1)**

UNIT2: General Principles

Concepts in discrete event simulation, Handling Stepped and Event-based Time in Simulations, Event scheduling/time advance algorithms, World views, List processing using dynamic allocation and linked list. **[4 hrs] (CO 1)**

UNIT 3: Statistical and Queuing Models in Simulation

Terms and concepts, Statistical models, Discrete and continuous distributions, Poisson distributions, Empirical distributions, Little's equation. Characteristics of queuing systems, Queuing notation, Long- Run measures of performance of queuing systems, Steady state behavior of infinite and finite calling population models, Use of network of queues.
[9 hrs] (CO 2)

UNIT 4 Random Number Generation

Pseudo random numbers, Techniques for generation of pseudo random numbers, Tests for random numbers, Random variate generation, Inverse Transform Technique- Exponential, Uniform, Weibull, Triangular distributions, Direct transformation for Normal and lognormal distributions. **[6hrs] (CO 2)**

UNIT 5 Input Modeling and Output Analysis of a Single Model

Data collection, Identifying the distribution of data - histograms and quantile plots, Parameter estimation, Goodness of fit tests applied to simulation inputs, Verification and validation of simulation models, Output analysis and measures of performance and estimation. **[6hrs] (CO 3)**

UNIT 6 Comparison and Evaluation of Alternative System Designs

Comparison of two system designs, Sampling with equal and unequal variances, Common random numbers, Comparison of several system designs, Linear regression, Random number assignment for regression. **[5 hrs] (CO 4)**

Course Outcomes:

After undergoing this course, the students will be able to

CO1: Discuss the fundamental elements of discrete-event simulation including statistical models, random processes, random variates, and inputs to simulation

CO2: Analyze a real world problem and apply modelling methodologies to develop a discrete-event simulation model

CO3 Interpret discrete-event techniques for solving a simulation problem

CO4: Compare and evaluate alternative system designs using sampling and regression

Suggested Readings/ Books:

1. Jerry Banks, John S. Carson II, Barry L.Nelson and David M.Nicol, "Discrete- event system and simulation", Prentice Hall of India.
2. Averill M.Law, "Simulation modeling and analysis (SIE)", Tata McGraw Hill India.
3. David Cloud, Larry Rainey, "Applied Modeling and Simulation", Tata McGraw Hill.
4. Gabriel A. Wainer, "Discrete-event modeling and simulation: a practitioner's approach", CRC Press.
5. Bernard P. Zeiger, Herbert Praehofer, Tag Gon Kim, "Theory of modeling and simulation: integrating discrete event and continuous complex dynamic systems", Academic Press.
6. Walter J. Karplus, George A. Bekey, Boris YakobKogan, "Modeling and simulation: theory and practice", Springer.

.----------

CourseCode: BTCS 607-18	**CourseTitle:** Simulation and Modeling Lab	**L:0;T:0; P:**	**Credits**

Detailed List of Tasks:

1. Implementation of Basic Operations on Matrices.
2. Implementation of Chi-square goodness-of-fit test.
3. Practical implementation of Queuing Models.
4. Design Inventory System.
5. Implementation of Monte-Carlo Simulation method.
6. Analysis of Discrete and Continuous Distributions.
7. Generation of Random Numbers using Linear Congruential Method.
8. Generation of Random Numbers using Combined Linear Congruential Method.
9. Evaluation of system design using Regression Analysis.
10. Simulate a network using any network simulator.

SuggestedTools - Scilab, Tortuga and Extend. Introduction to network simulators - NS2, CloudSim, Wireshark.